

NEWCASTLE GRAMMAR SCHOOL

Student Number: _____



2011 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

Examination Date: Friday 19th August

Examiner: Mr. M. Brain

General Instructions

- Reading Time - 5 minutes
- Working Time - 2 hours
- Write using a blue or black pen
- Write your student number on every booklet
- Board-approved calculators may be used
- A table of standard integrals is provided in this paper
- All necessary working should be shown in every question
- Each question attempted is to be returned in a separate Writing Booklet clearly marked Questions 1 etc.
- If required, additional booklets may be requested

Total marks - 84

- Attempt Questions 1- 7
- All questions are of equal value

Question 1 (Start a new booklet)**Marks**

- a) If $y = (\tan^{-1} x)^2$ find $\frac{dy}{dx}$ **2**
- b) Find the value of $\sum_{n=2}^5 {}^nC_2$ **2**
- c) Solve $\frac{2x-3}{x-2} \geq 1$ **4**
- d) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$ **1**
- e) Evaluate $\int_0^3 \frac{x}{\sqrt{1+x}} dx$ using the substitution $x = u^2 - 1$, where $u > 0$ **3**

Question 2 (Start a new booklet)

Marks

- a) Find the size of the acute angle between the two lines with equations $x + 2y + 1 = 0$ and $2x - 3y + 6 = 0$, correct to the nearest degree **4**
- b) A spherical balloon is expanding so that its volume, $V \text{ cm}^3$, increases at a constant rate of 72 cm^3 per second. What is the rate of increase of the surface area when the radius is 12 cm? **4**
- c) Find **4**
- (i) $\int \sin^2 x \, dx$
- (ii) $\int \frac{dx}{\sqrt{9 + 4x^2}}$

Question 3 (Start a new booklet)

Marks

- a) The rate of growth of a bacteria colony is proportional to the excess of the colony's population over 5000 and is given by

5

$$\frac{dN}{dt} = k(N - 5000)$$

- (i) Show that $N = 5000 + Ae^{kt}$ is a solution to the differential equation above
- (ii) Given that the initial population was 15000 and had risen to 20000 after 2 days find the value of A and k
- (iii) Hence calculate the expected population after 7 days

- b) The polynomial $P(x) = x^5 + mx^3 + nx$ has a remainder of 5 when divided by $(x - 2)$, where m and n are constants.

3

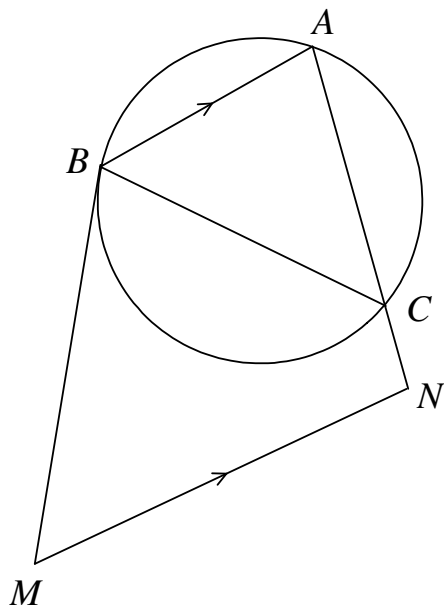
- (i) Prove that $P(x)$ is an odd function
- (ii) Hence find the remainder when $P(x)$ is divided by $(x + 2)$

- c) If α, β and γ are the roots of the equation $x^3 - 2x^2 + 3x + 7 = 0$ find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

4

- a) ABC is a triangle inscribed in a circle. M is a point on the tangent to the circle at B and N is a point on AC produced so that MN is parallel to BA

4



Copy the diagram into your answer booklet

- (i) State why $\angle MBC = \angle BAC$
- (ii) Prove that $MNCB$ is a cyclic quadrilateral
- b) Prove by mathematical induction that, for all integers $n \geq 1$

4

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n$$

- c) (i) Prove that the equation $\frac{\log_e x}{x} + 2 = 0$ has a solution between $x = 0.4$ and $x = 0.5$
- (ii) Use one application of Newton's method to find a closer approximation to the solution $x = 0.4$, correct to three decimal places

4

Question 5 (Start a new booklet)

Marks

- a) In Group A there are 5 men and 3 women. In Group B there are 4 men and 6 women. **4**
- (i) If one person is chosen at random from each group what is the probability that the two people chosen are of opposite sexes?
- (ii) If a group and then one person from that group is chosen at random what is the probability that the person chosen is a man?
- b) Using $\tan \frac{\theta}{2} = t$ show that $\frac{1 - \cos \theta}{\sin \theta} = t$ **3**
- c) A particle moves in a straight line such that its acceleration, a , is given by $a = 3x^2$, where x is displacement, v is velocity and t is time. Given that $v = -\sqrt{2}$ and $x = 1$ when $t = 0$ find x as a function of time, t **5**

Question 6 (Start a new booklet)

Marks

a) Consider the function $f(x) = 3 \sin^{-1}\left(\frac{x}{2}\right)$

5

- (i) State the domain and range of this function
- (ii) Sketch the graph of $y = f(x)$
- (iii) Find the slope of the graph at $x = 0$

b) The displacement, x metres, of a particle, at t seconds is given by:

7

$$x = 5 \cos(4\pi t)$$

- (i) Show that the acceleration of the particle can be expressed in the form:

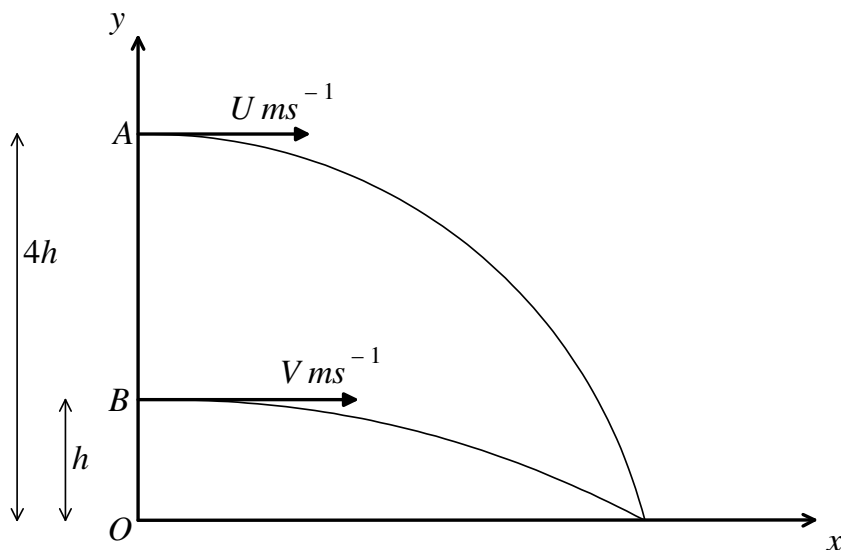
$$\ddot{x} = -n^2 x$$

- (ii) State the period, P , of the motion
- (iii) Determine the maximum velocity of the particle
- (iv) Determine the maximum acceleration of the particle
- (v) Express v^2 in terms of x , where v is the velocity of the particle

QUESTION 7 IS ON THE NEXT PAGE

- a) A vertical building stands with its base O on horizontal ground. A and B are two points on the building vertically above each other such that A is $4h$ metres above O and B is h metres above O . A particle is projected horizontally with speed U metres per second from A and 10 seconds later a second particle is projected horizontally with speed V metres per second from B . The two particles hit the ground at the same point and at the same time.

7



- (i) Show that the horizontal and vertical displacements of particles A and B , t seconds after the first particle is projected, are given by $x_A = Ut$, $y_A = 4h - \frac{1}{2}gt^2$ and $x_B = V(t-10)$, $y_B = h - \frac{1}{2}g(t-10)^2$, respectively
 - (ii) Find the time of flight of each particle
 - (iii) Prove that $V = 2U$
- b) The point $P(2t, t^2)$ is on the parabola with equation $x^2 = 4y$, having its focus at F . The point M divides the interval FP externally in the ratio 3:1.
- (i) Show that the co-ordinates of M are $x = 3t$ and $y = \frac{1}{2}(3t^2 - 1)$
 - (ii) Hence prove that the locus of M is also a parabola and determine the focal length of the locus of M

5

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

① a) $y = (\tan^{-1} x)^2$

$$\therefore \frac{dy}{dx} = 2(\tan^{-1} x)^1 \times \frac{d}{dx}(\tan^{-1} x)$$

$$= 2 \tan^{-1} x \times \frac{1}{1+x^2}$$

$$= \frac{2 \tan^{-1} x}{1+x^2} \quad (2)$$

b) $\sum_{n=2}^5 {}^nC_2 = {}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2$

$$= 1 + 3 + 6 + 10$$

(using calculator)

$$= (20) \quad (2)$$

c) $\frac{2x-3}{x-2} \geq 1$ x B.S. by (denom)²

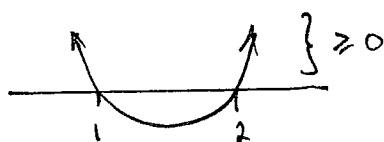
$$\therefore \frac{2x-3}{x-2} \times (x-2)^2 \geq 1 \times (x-2)^2$$

$$\therefore (2x-3)(x-2) \geq x^2 - 4x + 4$$

$$\therefore 2x^2 - 4x - 3x + 6 \geq x^2 - 4x + 4$$

$$\therefore x^2 - 3x + 2 \geq 0$$

Consider graph of $y = x^2 - 3x + 2$
 $= (x-1)(x-2)$



$$\therefore x \leq 1 \text{ or } x \geq 2$$

but $x \neq 2 \therefore x \leq 1 \text{ or } x > 2 \quad (4)$

d) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0} 3 \times \frac{\sin 3x}{3x}$

$$= 3 \times \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \quad \left. \begin{array}{l} \text{as } x \rightarrow 0 \\ 3x \rightarrow 0 \end{array} \right\}$$

$$= 3 \times 1$$

$$= (3) \quad (1)$$

e) $\int_0^3 \frac{x}{\sqrt{1+x}} dx$ $x = u^2 - 1$

$$\therefore dx = 2u du$$

$$= \int_1^2 \frac{u^2-1}{\sqrt{1+u^2-1}} \times 2u du \quad \text{at } x=3: u^2=4$$

$$u=2 (>0)$$

$$= \int_1^2 \frac{u^2-1}{u} \times 2u du \quad x=0: u^2=1$$

$$u=1 (>0)$$

$$= 2 \int_1^2 (u^2-1) du$$

$$= 2 \left[\frac{u^3}{3} - u \right]_1^2$$

$$= 2 \left(\left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right) \right)$$

$$= \left(\frac{8}{3} \right) \quad (3)$$

② a) $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$m_1: 2y = -x - 1 \therefore y = -\frac{1}{2}x - \frac{1}{2} : m_1 = -\frac{1}{2}$$

$$m_2: 3y = 2x + 6 \therefore y = \frac{2}{3}x + 2 : m_2 = \frac{2}{3}$$

$$\therefore \tan \theta = \left| \frac{-\frac{1}{2} - \frac{2}{3}}{1 - \frac{1}{2} \times \frac{2}{3}} \right|$$

$$= \left| -\frac{7}{4} \right|$$

$$= \frac{7}{4}$$

$$\therefore \theta = \tan^{-1} \left(\frac{7}{4} \right)$$

$$= 60.25 \dots$$

$$\theta = 60^\circ \quad (\text{nearest deg.}) \quad (4)$$

② b) S.A. = A, Volume = V

$$\therefore \frac{dA}{dt} = \frac{dA}{dV} \times \frac{dV}{dt} \dots \dots (1)$$

where $\frac{dV}{dt} = 72 \text{ cm}^3/\text{s} \dots \dots (2)$

Now: $\frac{dA}{dV} = \frac{dA}{dr} \times \frac{dr}{dV}$

where: $A = 4\pi r^2 \therefore \frac{dA}{dr} = 8\pi r \dots (3)$

$V = \frac{4}{3}\pi r^3 \therefore \frac{dV}{dr} = 4\pi r^2$

or $\frac{dr}{dV} = \frac{1}{4\pi r^2} \dots (4)$

$\therefore (2), (3), (4)$ into (1):

$$\begin{aligned} \frac{dA}{dt} &= 8\pi r \times \frac{1}{4\pi r^2} \times 72 \\ &= \frac{144}{r} \end{aligned}$$

$\therefore$ when $r = 12$

$$\begin{aligned} \frac{dA}{dr} &= \frac{144}{12} \\ &= \boxed{12 \text{ cm}^2/\text{s}} \quad (4) \end{aligned}$$

c) (i) $\int \sin^2 x \, dx = \frac{1}{2} \int 1 - \cos 2x \, dx$
 $= \frac{1}{2} (x - \frac{1}{2} \sin 2x) + C$

or $= \boxed{\frac{1}{2}x - \frac{1}{4} \sin 2x + C} \quad (2)$

(ii) $\int \frac{dx}{\sqrt{9+4x^2}} = \int \frac{1}{\sqrt{4(\frac{9}{4}+x^2)}} \, dx$

Using Standard Integrals sheet $= \frac{1}{2} \int \frac{1}{\sqrt{(\frac{3}{2})^2 + x^2}} \, dx$
 $= \boxed{\frac{1}{2} \ln(x + \sqrt{9/4 + x^2})} \quad (2)$

③ a) (i)

If $N = 5000 + Ae^{kt} \dots \dots (1)$

then $\frac{dN}{dt} = k \times Ae^{kt}$

but from (1): $Ae^{kt} = N - 5000$

$\therefore \frac{dN}{dt} = k(N - 5000)$

$\therefore N = 5000 + Ae^{kt}$ is a solution to $\frac{dN}{dt} = k(N - 5000)$ (QED) (1)

(ii) At $t=0$, $N = 15000$

$\therefore 15000 = 5000 + Ae^{k \times 0}$
 $= 5000 + A$

$\therefore \boxed{A = 10000}$

At $t=2$, $N = 20000$

$\therefore 20000 = 5000 + 10000e^{2k}$

$\therefore 10000e^{2k} = 15000$

$e^{2k} = 1.5$

$\ln(e^{2k}) = \ln 1.5$

$\therefore 2k = \ln 1.5$

$k = \ln 1.5 \div 2$

$= 0.20273 \dots$

$\boxed{k = 0.2027} \text{ (4dp)} \quad (3)$

(iii) $\therefore$ At $t=7$:

0.2027×7

$N = 5000 + 10000e$

$= 46325.7 \dots$

$\therefore \boxed{N = 46300} \text{ (nearest 100)} \quad (1)$

b) $P(x) = x^5 + mx^3 + nx$

(i) $\therefore P(a) = a^5 + ma^3 + na$

$$P(-a) = (-a)^5 + m(-a)^3 + n(-a)$$

$$= -a^5 - ma^3 - na$$

$$= -(a^5 + ma^3 + na)$$

$$\therefore P(-a) = -\{-(a^5 + ma^3 + na)\}$$

$$= a^5 + ma^3 + na$$

i.e. $P(a) = -P(-a) \therefore$ odd (QED) (1)

(ii) When $P(x)$ divided by $x+2$ remainder is $P(-2)$

AND when divided by $x-2$

remainder is $P(2) = 5$ (given)

Now, from (i) $P(a) = -P(-a)$

$$\therefore P(-2) = -P(-(-2))$$

$$= -P(2)$$

$$= -5 \quad (2)$$

c) α, β, γ : roots of $x^3 - 2x^2 + 3x + 7 = 0$

and: $\frac{1 \times \beta \gamma}{\alpha \times \beta \gamma} + \frac{1 \times \alpha \gamma}{\beta \times \alpha \gamma} + \frac{1 \times \alpha \beta}{\gamma \times \alpha \beta}$

$$= \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

Now: $\alpha\beta + \alpha\gamma + \beta\gamma = c/a$

$$= 3/1$$

$$= 3$$

$$\alpha\beta\gamma = -d/a$$

$$= -7/1$$

$$= -7$$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{3}{-7} \text{ or } -\frac{3}{7} \quad (4)$$

(4) a) (i) $\angle MBC = \angle BAC$

$\therefore$ angle between tangent and chord equals angle in alternate segment (1)

(ii) $\angle BAC + \angle CNM = 180^\circ$

(co-int. $\angle$'s in $\parallel$ lines)

and $\angle BAC = \angle MBC$ (from (i))

$$\therefore \angle MBC + \angle CNM = 180^\circ$$

$\therefore$ MNCB is cyclic quadrilateral

(opp. $\angle$'s add to 180°) (3)

b) Prove: $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n$
for $n \geq 1$

• Prove true for $n=1$

$$\text{i.e. LHS} = 1 \times 2^0 \quad \text{RHS} = 1 + (1-1) \times 2^1$$

$$= 1 \times 1 \quad = 1 + 0$$

$$= 1 \quad = 1 \quad (1/2)$$

$\therefore$ true for $n=1$

• Assume true for $n=k$

i.e. assume: $1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k$

$\therefore$ prove true for $n=k+1$

i.e. prove: $1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1)2^k = 1 + k \times 2^{k+1}$

Now: LHS = $1 + (k-1)2^k + (k+1)2^k$ {from assumption.

$$= 1 + k \times 2^k - 2^k + k \times 2^k + 2^k$$

$$= 1 + 2k \times 2^k$$

$$= 1 + k \times 2^1 \times 2^k$$

$$= 1 + k \times 2^{k+1}$$

$$= \text{RHS}$$

$$\therefore \text{LHS} = \text{RHS} \quad (3)$$

• $\therefore$ True for $n=1$ and true for

$n=k+1$ when true for $n=k$ (1/2)

$\therefore$ True for $n=1, 2, 3, \dots$ (all $n \geq 1$) (4)

(QED)

④ c) (i) Solution to $\frac{\log_e x}{x} + 2 = 0$
corresponds to $y = \frac{\log_e x}{x} + 2$ crossing
x-axis ($y=0$)

Now: when $x=0.4$:

$$\frac{\log_e x}{x} + 2 = \frac{\log_e 0.4}{0.4} + 2$$

$$= -0.29... \text{ (below x-axis)}$$

when $x=0.5$:

$$\frac{\log_e x}{x} + 2 = +0.61... \text{ (above x-axis)}$$

$\therefore$ Sign change $\Rightarrow$ solution between $x=0.4$ and $x=0.5$ (QED) ①

(ii) For $f(x) = \frac{\log_e x}{x} + 2$:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\text{where } f'(x) = \frac{x \times \frac{1}{x} - \log_e x \times 1}{x^2}$$

$$= \frac{1 - \log_e x}{x^2}$$

$$\therefore x_2 = 0.4 - \frac{\left(\frac{\log_e 0.4}{0.4} + 2\right)}{1 - \frac{\log_e 0.4}{(0.4)^2}}$$

$$= 0.4242...$$

$$\therefore \boxed{x_2 = 0.424} \text{ (3dp)} \quad \text{③}$$

⑤ a) $A = 5m, 3w$
 $B = 4m, 6w$

(i) $P(\text{opposite sexes}) = P(M, w \text{ or } w, M)$

$$= P(M, w) + P(w, M)$$

$$= \frac{5}{8} \times \frac{6}{10} + \frac{3}{8} \times \frac{4}{10}$$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $A \quad B \quad A \quad B$

$$= \boxed{\frac{21}{40}} \quad \text{②}$$

(ii) $P(\text{man}) = P(\text{Group A, M or Group B, M})$

$$= P(\text{Group A, M}) + P(\text{Group B, M})$$

$$= \frac{1}{2} \times \frac{5}{8} + \frac{1}{2} \times \frac{4}{10}$$

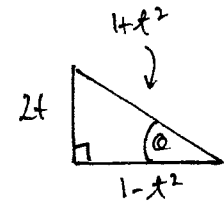
$$= \boxed{\frac{41}{80}} \quad \text{②}$$

b) $\tan \frac{a}{2} = t$

$$\therefore \tan(2 \times \frac{a}{2}) = \tan a$$

$$= \frac{2 \tan \frac{a}{2}}{1 - \tan^2 \frac{a}{2}}$$

$$= \frac{2t}{1-t^2}$$

$\therefore$  $\Rightarrow$

$$\sin a = \frac{2t}{1+t^2}$$

$$\cos a = \frac{1-t^2}{1+t^2}$$

$$\therefore \frac{1 - \cos a}{\sin a} = \frac{1 - \frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}} \times \frac{1+t^2}{1+t^2}$$

$$= \frac{1+t^2 - 1+t^2}{2t}$$

$$= \frac{2t^2}{2t}$$

$$= t$$

$\therefore$ LHS = RHS (QED) ③

$$(5) c) \frac{d}{dx}(\frac{1}{2}v^2) = a = 3x^2$$

$$\therefore \frac{1}{2}v^2 = \int 3x^2 dx$$

$$= x^3 + C$$

$$(C_1 = 2x^3)$$

$$\therefore v^2 = 2x^3 + C_1$$

$$\text{and at: } v = -\sqrt{2}, x = 1$$

$$\therefore 2 = 2 + C_1$$

$$\therefore C_1 = 0$$

$$\text{i.e. } v^2 = 2x^3$$

$$\therefore v = \pm \sqrt{2x^3}$$

$$\text{But initially } v = -\sqrt{2}$$

$$\therefore \text{select: } v = -\sqrt{2x^3}$$

$$\text{i.e. } \frac{dx}{dt} = -\sqrt{2} x^{3/2}$$

$$\therefore \frac{dt}{dx} = -\frac{1}{\sqrt{2}} x^{-3/2}$$

$$\therefore t = -\frac{1}{\sqrt{2}} \int x^{-3/2} dx$$

$$= -\frac{1}{\sqrt{2}} \times \frac{x^{-1/2}}{-1/2} + C_2$$

$$= \frac{2}{\sqrt{2}} x^{-1/2} + C_2$$

$$\text{i.e. } t = \sqrt{2} x^{-1/2} + C_2$$

$$\text{and at: } t = 0, x = 1$$

$$\therefore 0 = \sqrt{2} + C_2$$

$$\therefore C_2 = -\sqrt{2}$$

$$\therefore t = \frac{\sqrt{2}}{\sqrt{x}} - \sqrt{2}$$

$$\therefore \sqrt{x} = \frac{\sqrt{2}}{t + \sqrt{2}}$$

$$\therefore x = \frac{2}{(t + \sqrt{2})^2} \quad (5)$$

$$(6) a) (i) \text{ Domain:}$$

$$y = \sin^{-1} x : -1 \leq x \leq 1$$

$$\therefore y = \sin^{-1}(\frac{x}{2}) : -1 \leq \frac{x}{2} \leq 1 \quad (x^2)$$

$$\therefore -2 \leq x \leq 2$$

Range:

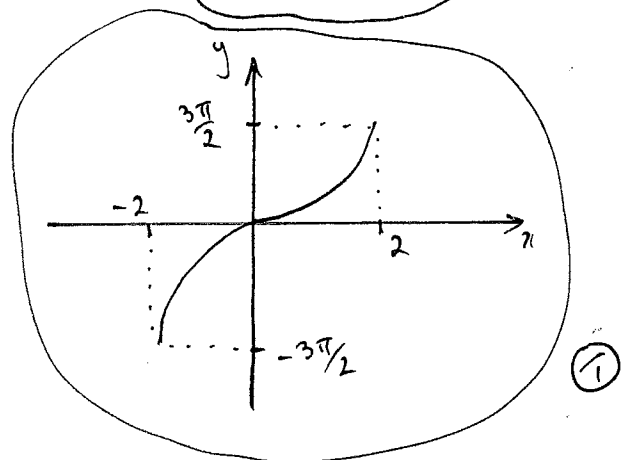
$$y = \sin^{-1} x : -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$y = 3\sin^{-1} x$$

$$\text{i.e. } \frac{y}{3} = \sin^{-1} x : -\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2} \quad (x^3)$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2} \quad (2)$$

(ii)



$$(iii) f(x) = 3\sin^{-1}(\frac{x}{2}) \quad \left\{ \begin{array}{l} \text{using} \\ \text{Standard} \\ \text{Integrals!} \end{array} \right.$$

$$\therefore f'(x) = 3 \times \frac{1}{\sqrt{2^2 - x^2}}$$

$$= \frac{3}{\sqrt{4 - x^2}}$$

$$\therefore \text{Slope at } x = 0 \text{ is } f'(0)$$

$$\text{and } f'(0) = \frac{3}{\sqrt{4 - 0^2}}$$

$$= \frac{3}{2} \quad (2)$$

⑥ b) $x = 5 \cos(4\pi t) \dots (1)$

(i) Express as $\ddot{x} = -n^2 x$

from (1): $\dot{x} = 5 \times -\sin(4\pi t) \times 4\pi$
 $= -20\pi \sin(4\pi t)$

$\therefore \ddot{x} = -20\pi \times \cos(4\pi t) \times 4\pi$
 $= -80\pi^2 \cos(4\pi t)$

$= -16\pi^2 (5 \cos(4\pi t))$ ②

i.e. $\ddot{x} = -(4\pi)^2 x = -n^2 x$ (QED)

(ii) Period $= 2\pi/n$

$= 2\pi/4\pi$

$= \frac{1}{2} \text{ (second)}$ ①

(iii) Max Velocity at centre i.e. at $x=0$

$v^2 = n^2(a^2 - x^2)$

where $n = 4\pi$, $a = 5$, $x = 0$

$\therefore v^2 = (4\pi)^2 (5^2 - 0)$

$\therefore v = \pm 4\pi \times 5$

$\therefore v_{\max} = 20\pi \text{ m/s}$ ②

(iv) Max. acceleration at amplitude (left)
 i.e. at $x=5$

$\therefore \ddot{x}_{\max} = -(4\pi)^2 \times 5$

$= -80\pi^2 \text{ m/s}^2$ ①

(v) from (iii) $v^2 = (4\pi)^2 (5^2 - x^2)$

$v^2 = 16\pi^2 (25 - x^2)$ ①

⑦ a)

(i) For 'any' projectile motion; fired from (0,0):

Horizontal

Vertical

$\ddot{x} = 0$

$\ddot{y} = -g$

$\therefore \dot{x} = C_1$

$\therefore \dot{y} = -gt + C_2$

but at $t=0$, $v_x = V \cos \theta$

but at $t=0$, $v_y = V \sin \theta$

$\therefore C_1 = V \cos \theta$

$\therefore C_2 = V \sin \theta$

$\therefore \dot{x} = V \cos \theta \dots (1)$

$\therefore \dot{y} = V \sin \theta - gt \dots (2)$

$\therefore x = Vt \cos \theta + C_3$

$\therefore y = Vt \sin \theta - \frac{1}{2}gt^2 + C_4$

but at $t=0$, $x=0$

but at $t=0$, $y=0$

$\therefore C_3 = 0$

$\therefore C_4 = 0$

$\therefore x = Vt \cos \theta \dots (3)$

$\therefore y = Vt \sin \theta - \frac{1}{2}gt^2 \dots (4)$

$\therefore$ in this question $\theta = 0$ for both projectiles

$\therefore \sin \theta = 0$, $\cos \theta = 1$

• for projectile A, time $= t$

$\therefore$ for projectile B, time $= t - 10$
 (10 seconds later)

• AND for projectile A:

equation (2) gives: $y = Vt \sin \theta - \frac{1}{2}gt^2 + C_4$

but at $t=0$, $y = 4h$ $\therefore C_4 = 4h$

for projectile B

at $t=0$, $y = h$ $\therefore C_4 = h$

$\therefore$ For A: (3) gives: $x_A = Ut$

'(4)' gives: $y_A = 4h - \frac{1}{2}gt^2$

For B: (3) gives: $x_B = V(t-10)$

$y_B = h - \frac{1}{2}g(t-10)^2$

(QED) ④

⑦(a) (ii) Time of flight : when $y=0$

ie. $y(A) = 0$

$y(B) = 0$

ie. $4h - \frac{1}{2}gt^2 = 0$

$\therefore h = \frac{1}{8}gt^2$ — (1)

and $h = \frac{1}{2}g(t-10)^2$ — (2)

(1) into (2) : $\frac{1}{8}gt^2 = \frac{1}{2}g(t-10)^2$

$\therefore t^2 = 4(t-10)^2$

$\therefore t = 2(t-10)$ or $-t = 2(t-10)$

$\therefore t = 2t - 20 \quad \therefore -t = 2t - 20$

$\therefore t = 20$ on $t = \frac{20}{3}$

(t = time of flight for A which must be > 10 seconds, for B to be projected 10 seconds after A)

$\therefore t = 20$ seconds

and $t-10 = 10$ seconds

$\therefore$ Time of flight : $\begin{matrix} A = 20 \text{ sec} \\ B = 10 \text{ sec} \end{matrix}$ ②

(iii) When both particles hit ground:

$x_A = x_B$

$\therefore 20u = v(20-10)$

$20u = 10v$

$\therefore v = 2u$ (QED) ①

b)(i) $x^2 = 4ay \rightarrow x^2 = 4y \quad \therefore a=1$

$\therefore$ focus is $(0,1)$, P is $(2t, t^2)$

Divide externally 3:1 $\left\{ \begin{array}{l} \therefore x = \frac{m x_2 + n x_1}{m+n} \\ \therefore m:n = 3:-1 \end{array} \right.$

$\therefore x = \frac{3 \times 2t - 1 \times 0}{3-1}$

ie. $x = 3t$ — (1)

and $y = \frac{m y_2 + n y_1}{m+n}$

$\therefore y = \frac{3 \times t^2 - 1 \times 1}{3-1}$

$y = \frac{3t^2 - 1}{2}$ — (2) (QED) ③

(ii) from (1) : $t = \frac{x}{3}$

sub into (2) : $y = 3\left(\frac{x}{3}\right)^2 - 1$

$= \frac{x^2}{3} - 1$

$\therefore 2y = \frac{x^2}{3} - 1$

$\therefore 6y = x^2 - 3$ (QED)

Equation of locus : $x^2 = 6y + 3$ ie parabola ①

and $x^2 = 6\left(y + \frac{1}{2}\right)$

$= 4 \times \frac{1}{2} \left(y + \frac{1}{2}\right)$

$\therefore$ focal length of

parabola is $\left(\frac{1}{2}\right)$ ①